

AKSHITA VERMA

(412) 251-2371 ◊ akshitav@andrew.cmu.edu ◊ [linkedin.com/in/akshita](https://www.linkedin.com/in/akshita) ◊ [github](https://github.com)

EDUCATION

Carnegie Mellon University

Master of Computational Data Science QPA: 4.00

Coursework - Machine Learning, Large Language Models Methods and Applications, Cloud Computing, Machine Learning in Production Foundations of Computational Data Science

Pittsburgh, PA

Dec 2026

Vellore Institute of Technology

Bachelor of Technology in Computer Science and Engineering

Vellore, India

July 2023

SKILLS

Programming & Systems: Python, Java, C++, SQL, JavaScript, TypeScript, Node.js, CI/CD, Git, SDLC

Cloud & Infrastructure:: AWS (EC2, S3, EKS), Azure, Google Cloud, Kubeflow, Docker, Kubernetes, SAP HANA Cloud

Application Development & DevOps: REST APIs, gRPC, React, MySQL, PostgreSQL, Terraform

Machine Learning & AI: PyTorch, Tensorflow/Keras, Pandas, NumPy, Scikit-learn, SciPy, LLMs, LoRA Fine-Tuning, CNNs, Hugging Face, Retrieval-Augmented-Generation (RAG), Natural Language Processing (NLP), Computer Vision

EXPERIENCE

Shell India Markets Private Limited

Associate Engineer

Associate Data Engineer

Bengaluru, India

Mar 2025 - July 2025

Aug 2023 - Feb 2025

- Developed **cloud-native microservices and REST APIs** (ReactJS, TypeScript, .NET, MySQL) for a document tracking system, deployed on Azure, **reducing emissions by 7%, saving \$4,000** annually, and earning the **Project Impact Award**.
- Designed a SQL data model in SAP HANA for financial reporting, integrating 100+ GB data; **cut memory usage by 85%** and **accelerated query runtime by 60%, saving \$130K**.
- Engineered reliable ETL processes by integrating Azure pipelines with Python automation for data cleaning, schema validation, and monitoring, enhancing data quality and **reducing refresh failures by 30%**.

Invigilo AI

AI Engineering Intern

Singapore, Singapore (Remote)

Oct 2022 - May 2023

- Engineered an end-to-end MLOps pipeline for YOLOv5, including COCO→YOLO annotation conversion, automated training, and deployment via Kubeflow + Kubernetes on AWS, achieving **98% hook block accuracy**, and **90% person accuracy**, **boosting efficiency by 20%**.
- Designed and built "Megatable," a centralized PostgreSQL-backed training data repository, **improved data accessibility by 5%** and **query performance by 40%** through a Streamlit-based UI.
- Automated data acquisition workflows by scripting ingestion pipelines across Google Drive and AWS S3, **reducing manual data handling by 60%** and **accelerating dataset preparation cycles by 30%** for the computer vision team.

PROJECTS

Carnegie Mellon University

Large-Scale Social Graph Ranking Service

Pittsburgh, PA

Mar 2026

- Built a distributed data pipeline using **PySpark** to process **1TB of social interaction data** on GCP, performing preprocessing, malformed JSON removal, and schema validation before loading into a structured **MySQL** database on **AWS EKS**, and implemented a composite ranking algorithm scoring users by interaction, keyword, and hashtag signals.
- Designed and deployed microservices for web authentication and storage using **gRPC** and **Kubernetes**, achieving **9k+ RPS throughput** with a ranked user interaction API optimized to **70ms latency** through query tuning and indexing strategies.

Carnegie Mellon University

Parameter-Efficient LLM Fine-Tuning with Tool Calling for Financial QA

Pittsburgh, PA

Nov 2025

- Designed and evaluated an end-to-end LLM pipeline on AWS using **PyTorch** and **Hugging Face Transformers**, comparing LoRA fine-tuning vs. in-context learning on **Llama-3-8B** and **Mistral-7B-Instruct-v0.2**; **achieved >2× accuracy improvement with 0.1% trainable parameters at ~\$3-4 compute cost**.
- Integrated tool-calling architecture for numeric execution, **improving execution accuracy from ~70-76% to 98-99%**, demonstrating scalable LLM system design for fintech applications.

National University of Singapore

Early Parkinson's Detection Tool

Singapore, Singapore

July 2022

- Designed and trained a CNN using **TensorFlow** and **OpenCV** for early Parkinson's disease detection on hand-drawn spiral images, engineering an end-to-end pipeline including data cleaning, normalization, and hyperparameter tuning to **achieve 96% classification accuracy**.
- Evaluated model performance using **Scikit-learn's metrics** suite, precision-recall, confusion matrix, and accuracy, and deployed results through an interactive Streamlit interface supporting real-time inference and experimental validation.

RESEARCH

Carnegie Mellon University

Data Influence on Vision Language Models

Pittsburgh, PA

Feb 2026 -Present

- Designing benchmark tasks and evaluation framework** for multimodal data influence in Vision-Language models (VLM) and Vision-Language-Action (VLA) models, investigating how gradient-based influence methods extend to joint visual-text representations and sequential decision-making for embodied control.
- Building an influence-based Retrieval Explorer** for scalable multimodal data curation, enabling interactive analysis of training data contributions to cross-modal alignment, grounding robustness, and visual reasoning across web-scale image-text corpora.